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BIOLOGY
STANDARD LEVEL
PAPER 2

SPECIMEN PAPER

1 hour 15 minutes

Candidate session number

Examination code

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INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all questions.
- Section B: answer one question.
- Write your answers in the boxes provided.
- A calculator is required for this paper.
- The maximum mark for this examination paper is *[50 marks]*.

20 pages

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SECTION A

*Answer **all** questions. Write your answers in the boxes provided.*

1. Large areas in northern Indiana (USA) are used for the cultivation of corn (*Zea mays*). Headwater streams are water bodies that drain the land and merge to form larger rivers.

Ecologists asked questions after the introduction of genetically modified *Bt* corn varieties:

- does organic debris from fields of genetically modified corn enter headwater streams?
- are there any effects on stream life?

To try to answer these questions, a group of researchers took measurements from twelve headwater streams in Indiana and carried out laboratory experiments. Corn crops were growing on both sides of each of the streams.

After the corn was harvested, traps were used to measure the amount of corn leaves, cobs and pollen entering the streams. The results are shown in the table.

Stream	**Leaf and cob inputs / $g\,m^{-2}\,y^{-1}$**	**Pollen inputs / $g\,m^{-2}\,y^{-1}$**
1A	0.1	0.03
1B	1.3	0.73
1C	0.7	0.09
1D	3.4	0.39
1E	1.5	0.59
1F	8.2	0.16
2A	3.0	1.05
2B	0.0	0.27
2C	0.2	0.37
2D	0.3	0.24
2E	0.6	0.31
2F	0.9	0.38

[Source: adapted from Rosi-Marshall, *et al.*, (2007), *Proceedings of the National Academy of Sciences*, **104**, pages 16204–16208.
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(Question 1 continued)

(a) Identify the stream with the highest total input of organic debris from corn crops. *[1]*

. . . . . .

(b) Determine, with a reason, whether the data in the table supports the hypothesis that the inputs of leaves and cobs to streams is always greater than the input of pollen. *[1]*

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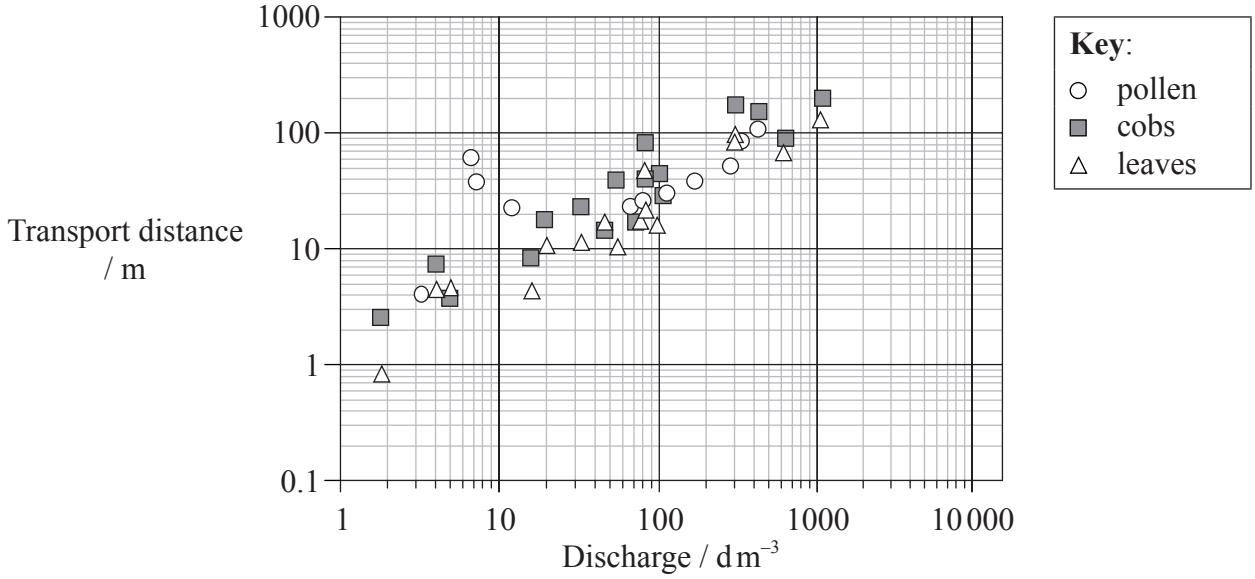
(Question 1 continued)

Leaves, cobs and pollen were stained with a pigment and then released into the streams to find the average transport distance. The volume of water flowing in the streams (discharge) was also measured. The results are shown in the graph.



[Source: adapted from Rosi-Marshall, *et al.*, (2007), *Proceedings of the National Academy of Sciences*, **104**, pages 16204–16208. Copyright (2007) National Academy of Sciences, U.S.A.]

(c) Estimate the maximum transport distance for cobs. *[1]*

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(d) State the relationship between discharge rates and transport distance for debris from the corn crop. *[1]*

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(Question 1 continued)

(e) Distinguish between the transport distance of cobs and leaves. *[2]*

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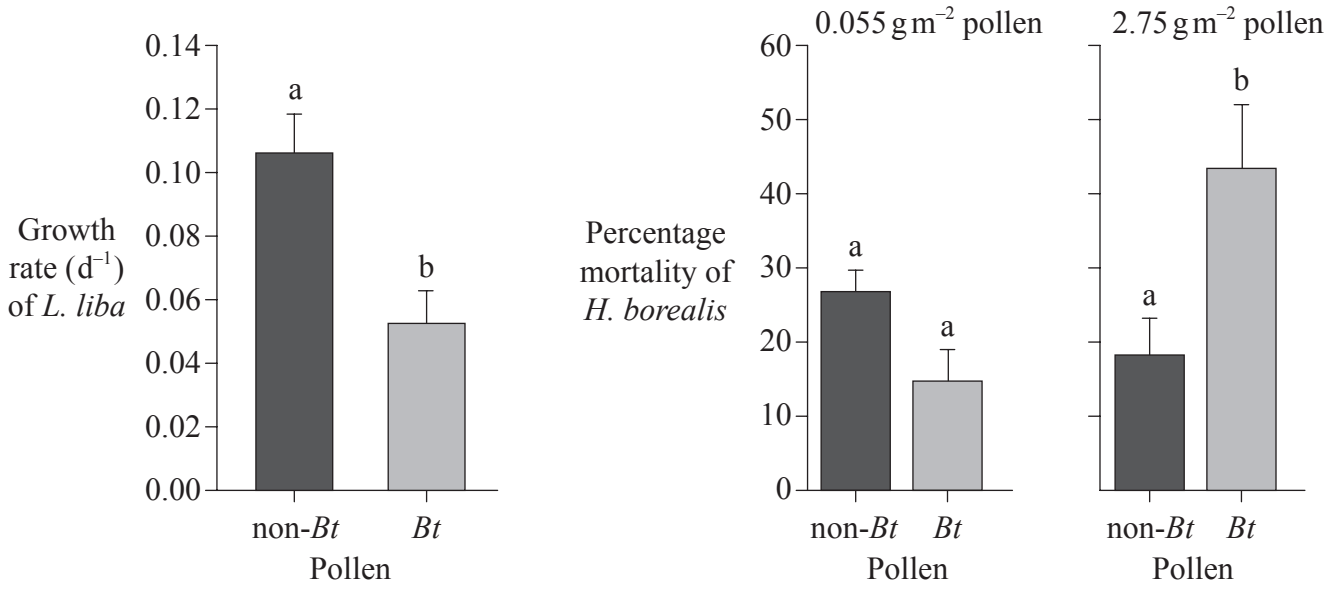
(Question 1 continued)

Larvae of caddis flies were observed in parts of the headwater streams where organic debris from corn crops had accumulated. Laboratory experiments were performed to try to find out if these larvae could be affected by consuming debris from *Bt* corn.

Lepidostoma liba is a caddis fly that feeds by shredding dead plant matter. The growth rate of *L. liba* was measured when it was fed on leaves from *Bt* corn and non-*Bt* corn. *Helicopsyche borealis* is a caddis fly that feeds by scraping algae surfaces. The mortality rate of *H. borealis* was measured when it fed on biofilms of algae containing *Bt* pollen or non-*Bt* pollen.

Two concentrations of pollen were tested. One concentration ($0.055\,g\,m^{-2}$) was based on the maximum observed daily pollen input rate to streams. The other concentration was 50 times higher ($2.75\,g\,m^{-2}$).

The results are shown in the bar charts. Statistically significant differences between non-*Bt* pollen and *Bt* pollen in each bar chart are indicated by different letters on error bars.



[Source: adapted from Rosi-Marshall, *et al.*, (2007), *Proceedings of the National Academy of Sciences*, **104**, pages 16204–16208. Copyright (2007) National Academy of Sciences, U.S.A.]

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(Question 1 continued)

(f) Using the data in the bar charts, evaluate the effects of the *Bt* pollen on caddis flies. *[3]*

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The research described in this question was strongly criticized soon after it was published by some other biologists and by the company that produced *Bt* varieties of crop plants. In particular there were objections to a statement in the research paper that "Widespread planting of *Bt* crops has unexpected ecosystem-scale consequences".

(g) Discuss whether this statement in the research paper was justified, based on the methods used in the research and the data obtained. *[3]*

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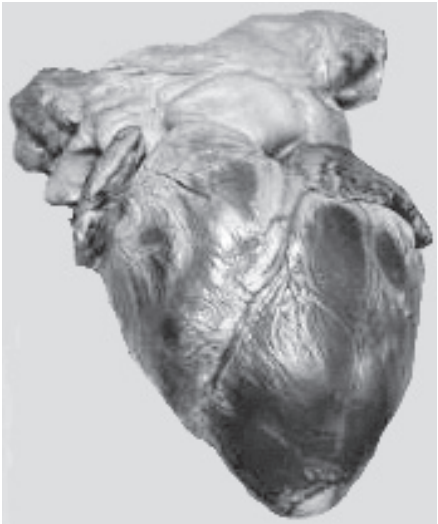
2. The photograph shows a heart, viewed from the ventral side.



(a) Label the photograph to show the position of

- the right ventricle
- the left atrium
- a coronary blood vessel. *[3]*

(b) Outline **two** ways the body uses to increase the heart rate. *[2]*

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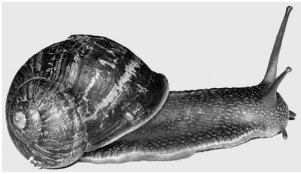
Please **do not** write on this page.

Answers written on this page
will not be marked.

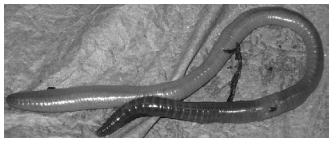
3. The photographs show different organisms (not to scale).

Hadronyche modesta

Cardisoma armatum



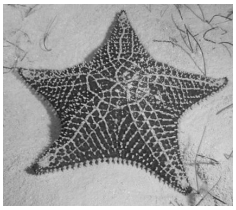
Helix aspersa



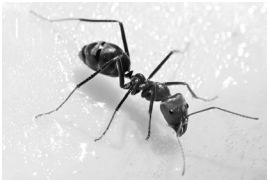
Lumbricus terrestris

Pteridium aquilinum

Zalophus wollebaeki



Oreaster reticulatus



Iridomyrmex purpureus

Helianthus annuus

(a) Deduce all of the organisms that belong to each of the following phyla. *[3]*

Filicinophyta:

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Arthropoda:

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Mollusca:

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(Question 3 continued)

(b) Explain how scientists avoid confusion over the names of living organisms. *[3]*

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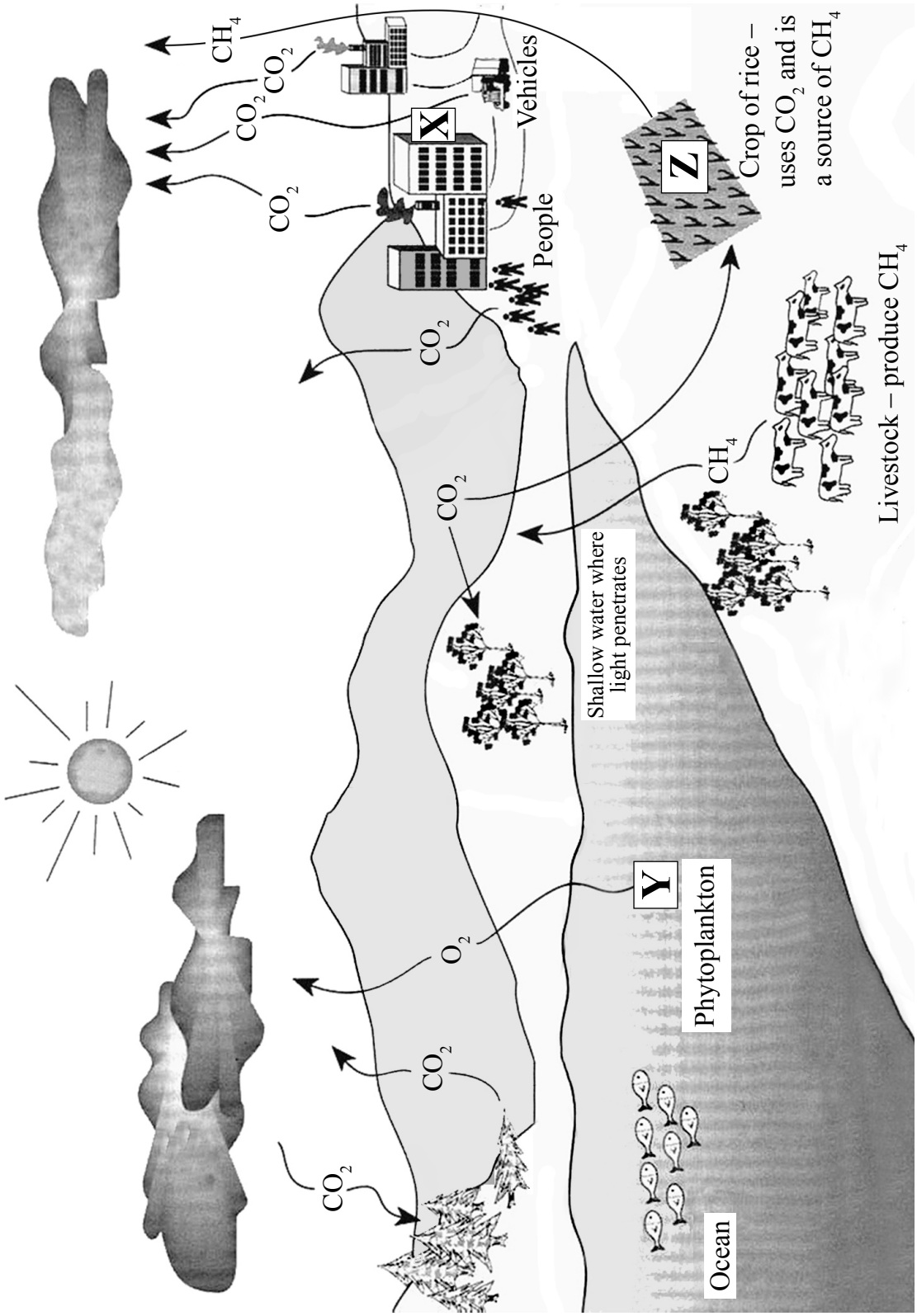
4. The diagram shows the carbon cycle.



[Source: adapted from www-das.uwyo.edu/~geerts/cwx/notes/chap01/carbon_cycle.jpeg]

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(Question 4 continued)

(a) State the processes occurring at X and Y. *[2]*

X: . . . . . .

Y: . . . . . .

(b) Predict the conditions that would increase the release of methane shown at Z. *[2]*

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(c) Outline the impact of the gases shown in the diagram on the greenhouse effect. *[2]*

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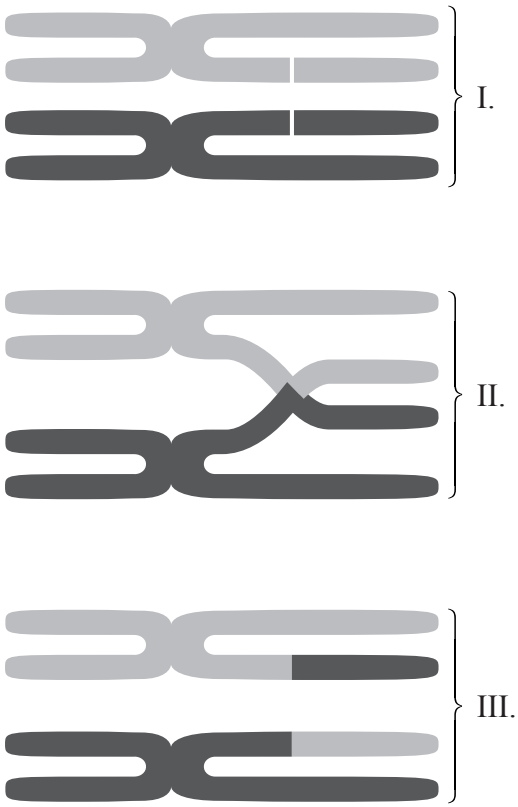
5. The diagrams show two chromosomes at three stages in meiosis.



(a) The diagrams show a process that promotes genetic variation. Describe this process. *[3]*

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(Question 5 continued)

(b) State the type of life-cycle that includes meiosis and the reason for it being needed in this type of life-cycle. *[2]*

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SECTION B

*Answer **one** question. Up to [1] additional mark is available for the quality of your answer.*

6. Antibiotics can be used to treat bacterial infections in human tissues because of differences in cell structure between prokaryotes and eukaryotes.

(a) Distinguish between the structure of prokaryotes and eukaryotes. *[8]*

(b) Evaluate the drug tests that Florey and Chain carried out on penicillin. *[3]*

(c) Explain the reasons for the ineffectiveness of antibiotics in the treatment of viral diseases. *[4]*

7. Humans need a supply of energy for processes such as active transport in cells.

(a) Explain how humans release energy from digested foods to make it available for processes in cells. *[7]*

(b) Describe **one** example that occurs in axons for each of the following

- active transport
- and facilitated diffusion. *[5]*

(c) Outline how biologists can ensure that research into energy release involving animals is ethically acceptable. *[3]*